

Talha Bilal

Senior Software Engineer & Technical Team Lead

Java | Spring Boot | Keycloak | AD/LDAP | OIDC SSO | PKI | HSM

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Pakistan (UTC+5) | English - professional | Full-time, contract & projects

Specializations: Senior Java / Spring Boot | Backend / Platform | IAM & PKI / Security Engineering

Senior Java / Spring Boot engineer & technical lead - on-prem Keycloak (AD/LDAP federation, OIDC application SSO, custom SPIs), plus PKI, CSC remote signing, Java Card, and HSM platforms. Ships secure microservices and REST APIs for regulated, multi-tenant environments.

KEY STRENGTHS

- Java 17/21 & Spring Boot microservices - REST/OpenAPI, Spring Security, event-driven (RabbitMQ)
- On-prem Keycloak - AD/LDAP federation, OIDC SSO, realm/client policies, custom SPIs
- PKI & digital signing - CA, OCSP/CRL, CSC remote signing, S/MIME, RFC 3161 timestamping
- HSM integration - PKCS#11, AWS CloudHSM, session pooling and production hardening
- Team leadership - architecture, code review, CI/CD, on-call runbooks in audit-sensitive domains

CORE SKILLS

Java 21/17, Spring Boot, Spring Security, REST / OpenAPI, Microservices, Event-driven (RabbitMQ), PKI / CA, OCSP / CRL, S/MIME & CMS, CSC API (remote signing), ETSI EN 319 401 / 411, PKCS#11, Keycloak (on-prem), LDAP / Active Directory federation, OAuth 2.0 / OIDC, Application SSO integration, Keycloak SPI (Authenticator, User Storage, Protocol Mapper), Bouncy Castle, IAIK, RFC 3161 TSA, Java Card 3.x, GlobalPlatform 2.x, APDU / host tools, AWS CloudHSM, Utimaco, SoftHSM2, Docker, CI/CD (Jenkins, GitHub Actions), PostgreSQL / MySQL, Redis, Integration & API testing, Structured logging & production support

EXPERIENCE

Dictalabs - Technical Team Lead - Java Spring Boot & PKI

Aug 2024 - Present | Enterprise digital trust & signing (B2B PKI platform)

- Lead engineers owning microservices for PKI, CMS/PKCS#7 signing, and HSM-backed key operations across production tenants.
- Architected PKCS#11 integration layer (AWS CloudHSM, SoftHSM2) - improved signing stability via pooled sessions, idempotent retries, and operational runbooks.
- Built CSC-compliant remote signing service (credentials, info, signatures, timestamps) with OAuth-secured API access and HSM-backed signature creation for client apps.
- Delivered on-prem Keycloak in production: LDAP/Active Directory user federation, group-to-role mapping, OIDC clients for application SSO, realm policies, and custom SPIs for certificate-aware login where required.
- Ship Java Card 3.x applets (RSA/ECDSA on-card) with GlobalPlatform lifecycle and host issuance tooling.
- Drive RFC 3161 timestamping, OCSP/CRL hardening, and incident response with security and product leadership.

Codegic - Senior Software Engineer - Java Spring Boot & PKI

Aug 2021 - Jul 2024 | PKI & secure document/signing products

- Built production CA stack: issuance, renewal, OCSP/CRL responders, and tenant-scoped certificate policies.
- Developed S/MIME/CMS email signing and Crypto-Core services integrated with enterprise HSMs and audit trails.
- Implemented Java Card applets and PC/SC host applications for smart-card issuance and secure key ceremonies.
- Established Docker-based CI/CD, API regression suites, and performance tuning for high-volume signing paths.
- Delivered multi-tenant PDF/signing workflows with approval chains and compliance-oriented logging.

SELECTED PROJECTS

- Keycloak IAM - on-prem, AD/LDAP federation, OIDC SSO, custom SPIs

- CSC Remote Signing - credentials, signHash, OAuth, HSM-backed signatures
- PKI Server - CA, OCSP/CRL, HSM-backed multi-tenant issuance
- PKCS#11 Service - pooled HSM sessions, multi-vendor support
- Java Card - GP 2.x lifecycle, RSA/ECDSA applets, host integration
- Secure Email Signing - S/MIME/CMS, RSA & ECDSA
- Workflow Management - PDF signing, approvals, audit trails

HIGHLIGHTS

- On-prem Keycloak with Active Directory & application SSO: Employees sign in once with AD credentials and access integrated apps via SSO; IT has one IdP to manage users, clients, and policies instead of scattered login implementations.
- Keycloak IAM with PKI-backed authentication: Single IAM layer for admin and customer apps, reduced bespoke auth code in services, and auditable login flows aligned with PKI issuance.

EDUCATION

B.S. Computer Science - KFUEIT, 2021